



Pitwall

1. Introduction

Pit Wall is a prototype fan-engagement application built for the Cognizant × Aston Martin Aramco Formula One Team Ideathon — a collaborative innovation event where teams designed new digital experiences for the Aston Martin F1 fanbase. Unlike a full production build, Pit Wall was created in the compressed timeframe of an ideathon and is intended as a concept prototype that demonstrates the idea, the user journey, and the interface, rather than a finished, backend-connected product.

The project addresses a gap in how Formula One fans experience race strategy. Strategy is one of the most decisive and intellectually engaging parts of an F1 race, yet for fans it is almost entirely passive — decisions happen on the pit wall and are only explained afterwards. Pit Wall reframes this by putting the fan on the Aston Martin pit wall: during a race, the same strategic decisions the real team faces are surfaced to the user as timed "make the call" prompts, supported by an AI race engineer that explains the situation in plain language and is scored against the team's real-world calls.

The system is built as a cross-platform Expo / React Native application that runs on iOS, Android, and the web. For the ideathon it is primarily demonstrated on the web inside a phone-style frame. Because it is a prototype, all data is held in memory during a session and the "AI race engineer" is simulated through scripted, rule-based responses that stand in for a live language model. This keeps the prototype self-contained and fully demonstrable without needing live race feeds, a backend, or paid API access.

1.5 Project Function and What It Does

Pit Wall is an interactive F1 strategy companion that lets a fan take control of an Aston Martin driver for a race and make the strategy calls themselves. From a user's point of view, the app first personalises itself through onboarding — the user creates an account, answers a short quiz that sets their "fan archetype" (from New Fan to Super Fan), and customises their profile. This archetype then adapts the whole experience: how much data is shown, how much guidance is given, and how technical the explanations are.

Before a race, the user selects which Aston Martin driver to control and locks in pre-race predictions such as pit stops, tyre stints, and pace. During the race, timed decision cards appear (for example, whether to box under a safety car, or whether to deploy stored energy — a 2026-rules-inspired call). The user can ask the AI race engineer for insight, which responds at the appropriate level for their persona, and then commits to a call under a countdown. Afterwards, a post-race debrief scores the user's decisions against Aston Martin's real strategy — a recurring

"Beat Aston Martin" theme — and adds points to their season total. Fans can also explore the season calendar and standings, compete on global and private-league leaderboards, and manage their settings. Overall, the core function of the prototype is to turn passive strategy-watching into an active, personalised, and competitive fan experience.

2. Tech Stack Used

- **TypeScript**
- **React**
- **React Native**
- **Expo (SDK 56)**
- **react-native-web**
- **react-native-svg**
- **react-native-safe-area-context**
- **@expo/vector-icons (Feather & Material Community Icons)**
- **expo-status-bar / expo-font**

Expo and React Native were chosen so that a single TypeScript codebase could run on iOS, Android, and the web, which suited an ideathon where the idea needed to be demonstrated quickly and on whatever device was available. React Native's component model made it possible to build a polished, app-like interface at speed, while react-native-web allowed the same app to be shown in a browser inside a phone frame during pitching.

react-native-svg was used to draw the custom visual elements that give the prototype its identity — the live circuit map with an animated car marker, the tyre and telemetry diagrams, and the "liquid glass" ambient gradient background used for the Aston Martin-inspired look.

@expo/vector-icons supplied a consistent icon set across every screen, and react-native-safe-area-context handled device insets (notch and dynamic island) so the layout felt native.

Because Pit Wall is a prototype, it deliberately does not include a backend, database, or authentication server. Account details, customisation choices, race decisions, and scores are held in in-memory React state for the duration of a session rather than persisted. The AI race engineer is simulated — its responses are produced by scripted, persona-aware logic rather than a live AI model — which lets the prototype demonstrate exactly how an AI-assisted strategy experience would feel while remaining fully self-contained and offline. In a production version, these mock layers would be replaced by real services (a live language model for the engineer, a data provider for timing/telemetry, and cloud storage with authentication for accounts and progress). Real Formula One team logos were intentionally avoided for copyright reasons and replaced with stylised team-colour badges..

3. Architecture Used

Pit Wall follows a lightweight, component- and context-based architecture appropriate to a prototype. Rather than a large framework, the app is organised around a single top-level component (PitWallApp) that owns the global state and composes the individual screens, with shared state distributed through React Context providers. This keeps the concept easy to reason about and fast to iterate on during an ideathon.

State is separated by concern into distinct contexts: a Fan context holds the user's persona (New / Casual / Super / Technical) and drives how every screen adapts its content; a Game context holds the active race entry (chosen driver, stint plan, pace) and the user's live decisions; a Customisation context holds profile cosmetics (name, avatar, colour scheme, banner); and a UI context handles global toasts and confirmation dialogs. A helper (byLevel) is used throughout so that text, data richness, and guidance can be selected per persona from one place.

Navigation is handled by a simple internal model — a current tab plus a screen "stack" for detail views — rather than a heavyweight router, which is sufficient for a prototype and keeps the whole flow inspectable. On the web, an outer shell (App.tsx) renders the app inside a scaled phone frame so it presents like a mobile device during demos. Visual identity is centralised in a single design system: an Aston Martin brand token set (deep racing green, lime accent, translucent glass surfaces) and one shared stylesheet, so the futuristic "liquid glass" theme applies consistently across every screen.

4. Full Functionality with Technical Implementation

The application starts in App.tsx, which wraps the prototype in a safe-area provider and, on the web, places it inside a phone-style frame with an ambient Aston Martin-green backdrop. Control then passes to PitWallApp, which decides whether to show onboarding or the main interface based on whether a fan persona has been set for the session. Because the prototype has no persistence, every launch begins as a fresh first-run, which is ideal for repeatable demonstrations.

The onboarding flow establishes the personalised experience that the rest of the app depends on. It begins with account creation — name, email, and password, plus a profile icon — framed as a single login that would, in a full build, work across devices and a future web version. The user then completes a short quiz that determines their fan archetype (New Fan through Super Fan), which is the foundation of personalisation throughout the app. Next comes profile customisation (banner, colour scheme, and earned badges), which also introduces a "Pit Wall Pro" subscription concept as a future monetisation avenue. Onboarding then presents a short tutorial, an explicit AI transparency screen stating that the AI and live data may not always be fully accurate, and finally a separate marketing-consent step. This ordering was a deliberate design choice: account first (for identity), then personalisation, then responsible disclosure and consent.

After onboarding, the user lands in the main navigation shell with tabs for Home, Season, Live, Leagues, and Settings. The Home dashboard summarises the fan's state — the next session, an AI

pre-race briefing (whose wording adapts to persona), their last race debrief, and shortcuts into race setup. The Season section provides the full calendar with past and upcoming rounds, drill-down race pages with session results, and championship standings that can be toggled between Drivers and Constructors.

The race setup flow is where the fan takes control. They choose which Aston Martin driver to run (Alonso or Stroll, carrying a loyalty bonus) and lock in a pre-race strategy of planned stops, tyre stints, and pace. This entry is stored in the Game context so it carries through to the live race, where the chosen driver is highlighted on the circuit.

The Live race screen is the core of the prototype. A custom SVG circuit map shows the field with the controlled driver marked, alongside a live leaderboard. A Race Data telemetry sheet presents tyre wear, energy deployment, gap deltas, and a weather timeline — and this is where the persona system is most visible: a Super Fan sees full deep-dive telemetry and diagrams, whereas a New Fan sees the same underlying data as simplified, plain-language explanations. An AI Radio panel lets the fan message the AI race engineer in real time (for example, "Where is Russell?") and receive contextual, persona-appropriate replies. The heart of the loop is the timed decision card: when Aston Martin faces a strategic moment, the same call is pushed to the user with a countdown. They can tap "Ask the Engineer," which shows the engineer's reasoning together with a confidence rating — and when confidence is low, a flashing red "AI discrepancy" warning reinforces the transparency message that the AI can be wrong before the user commits.

Once the race calls are made, the post-race debrief scores the user. A scoring engine grades each decision (strategy quality weighted against outcome, with a risk multiplier that rewards bold correct calls and penalises bold wrong ones), compares the user directly against Aston Martin's real strategy, and rolls the result into a season total with tiers from Rookie up to Pit Wall Master. This turns each race into a measurable, competitive result rather than a one-off interaction.

The remaining sections round out the experience. Leagues provide global and private-league leaderboards so fans can compete against the wider Aston Martin fanbase or against friends and family. Settings let the user switch their fan persona, adjust AI and data preferences, and view a data/privacy note, and the customised profile (avatar, banner, colours) is reflected consistently across the interface. Finally, the entire prototype was given a futuristic "liquid glass" visual treatment with Aston Martin branding — translucent glass surfaces over an ambient racing-green glow, hairline light edges, and lime accent glows — to make it feel like a premium, modern team product.

As a closing note, it is worth restating that Pit Wall is a prototype produced for the Cognizant × Aston Martin Formula One Team Ideathon. Its purpose is to prove out the concept and the fan experience end-to-end; the account system, data feeds, and AI engineer are simulated stand-ins that demonstrate how the finished product would behave, and would be replaced with real authentication, live timing data, and a genuine AI model in a production version.